

I build and evaluate AI systems around how people think, decide and interact. My work combines machine learning, behavioural science and human-computer interaction. At ETH Zürich, I teach across human-computer interaction, user experience, human cognition and AI's role in society.

Academic & Research Experience

- Dec 2023–Present **Senior Researcher & Lecturer, Applied Data Science in Cognitive Architecture, Chair of Cognitive Science, ETH Zürich & Singapore-ETH Centre (SEC/FCL), Zurich / Singapore**    
- **Associated Researcher, ETH AI Center.** Lead the AI research line of the Chair of Cognitive Science; build and lead interdisciplinary AI / data-science / architecture teams across hubs.
 - **Human-AI Augmentation:** design and study AI support systems for human cognition, leading the *Intelligent Decision Assistant for Safe Drug Management* in complex poly-pharmacy / multi-morbidity patients (industry partners & University Hospital Zurich).
 - **Cognitive Primitives:** externalized, auditable procedural memory for OOD generalization in language models (GPU-validated design).
 - **Evidence-based hospital emergency departments:** non-intrusive wearable IoT proximity sensors + ML to map staff/patient movement onto architectural layouts.
 - **Holistic tissue analysis:** AI/ML for 3D non-invasive histopathology of thyroid carcinomas for precision medicine.
 - **Visiting Scholar,** Cognitive Architecture group, University of Cambridge (Feb–Apr 2025).
- Apr 2019–Dec 2023 **Lecturer & Senior Researcher (2020–2023); Postdoctoral Researcher (2019–2020), Data Science, Systems & Services (DS3) & Professorship of Cognitive Science, ETH Zürich, Zurich, Switzerland** 
- **Built and evaluated AI-supported parenting tools** for rural Peru as part of the [Digital Support Systems](#) team; contributed software, data curation and validation to a cluster-randomized trial of **2,461 caregiver-child pairs** (*Science Advances*, 2026).
 - Led **MLOps / AutoML** infrastructure: terabyte-scale pipelines (MeteoSwiss), integration with reproducible data pipelines (SDSC / Renku), and the [Ease.ML lifecycle/CI platform](#) (CIDR 2021).
 - Built AI/ML systems for **3D non-invasive histopathology** of thyroid tumors (precision medicine); advanced **agent-based models** in navigation, contrasting VR experiments with human behaviour.
- Apr 2017–Mar 2019 **Lecturer & Postdoctoral Researcher, Professorship of Computational Social Science (COSS), ETH Zürich, Zurich, Switzerland** 
- Enhanced **agent-based models** for large-scale simulation: synthetic populations of deep / shallow reinforcement-learning agents studied in evolutionary and evacuation contexts.
 - Taught *Modeling and Simulating Social Systems* (851-0585-04L, 860-0011-00L).
 - Preceded by a **Visiting Researcher** stay at COSS (Apr–Jun 2016), prototyping deep-RL agents in an evolutionary agent-based model.
- Apr 2015–Mar 2017 **Postdoctoral Researcher, Centre for Large-scale Earthquake, Tsunami & Disaster, Earthquake Research Institute, The University of Tokyo, Tokyo, Japan** 
- **First author** of a hybrid MPI+OpenMP HPC multi-agent evacuation simulator (**2M agents, 81 km²**) on Japan's **K computer: 75.9% strong scaling to 2,048 cores**; framework later [scaled to 10M agents / 16,384 cores](#).

Education

- 2012–2015 **Ph.D., Computational Science & High-Performance Computing, The University of Tokyo, Japan** 
- 2009–2011 **M.Sc. in Road Engineering (Magna Cum Laude), Universidad de San Carlos de Guatemala, Guatemala** 
- 2004–2008 **Civil Engineer, B.Sc. (Cum Laude), Universidad del Valle de Guatemala, Guatemala** 

Technical Skills

- Programming Python, C++, Java, JavaScript, R, MATLAB/Scilab, SQL, VBA; Maxima, Sage, Mathcad.
- HPC & Parallel MPI (distributed memory), OpenMP (shared memory), CUDA-C (GPU); large-scale runs on the K computer.

AI / ML Machine learning, deep & shallow reinforcement learning, retrieval-augmented LLMs, transfer learning; AutoML / MLOps.

Methods Agent-based modelling & large-scale simulation, numerical analysis, multi-agent systems.

Tools & Data Git/GitHub; IoT & proximity sensors, LiDAR; GIS (OpenStreetMap/Osmosis), Google Maps API.

Languages

Spanish Native.

English Full professional proficiency (C1).

German Full professional proficiency (C1); *Deutsches Sprachdiplom Zweite Stufe*.

Japanese Basic (A1).

Teaching at ETH Zürich

2024–2026 **Human-Computer Interaction: Cognition and Usability** (851-0252-01L). Teaching team.

2025–2026 **Quantitative User Experience Research: Surveys and Logs Analysis** (851-0257-00L). Teaching team.

2026 **What is ‘intelligence’ - Human Cognition and Artificial Intelligence** (851-0259-00L). Teaching team.

2025 **Pandora’s Box: Large Language Models and their Impact on Society** (851-0258-00L). Teaching team.

2024–2025 **Evidence-Based Design: Methods and Tools for Evaluating Architectural Design** (851-0252-08L). Teaching team.

2026 **Advanced Topics in Evidence-Based Design for Architecture** (851-0253-08L). Teaching team.

2025 **Joint doctorate ETH-EPFL: Introduction to Methods in Learning Sciences II** (851-0255-00L). Teaching team.

2024 **Joint doctorate ETH-EPFL: Introduction to Methods in Learning Sciences I** (851-0252-14L). Teaching team.

2026 **Research Seminar Cognitive Science** (851-0252-05L). Teaching team.

2026 **AI4GOOD: Human-Centered AI for Social Good: Peace, Health, Climate**. Teaching team.

2018 **Lecture with Computer Exercises: Modeling and Simulating Social Systems in MATLAB (or Python)** (851-0585-04L). Teaching team.

2018 **Modeling and Simulating Social Systems in MATLAB (or Python) - With Coding Project** (860-0011-00L). Teaching team.

Selected Publications

2026 **Aguilar, L.**, Nagler, J., Hoelscher, C., Antulov-Fantulin, N. *Does Your Neural Network Extrapolate? Feature Engineering as Identifiability Bias for OOD Generalization*. Advances in Neural Information Processing Systems (NeurIPS) (accepted; poster).

2026 Fink, G., McCoy, D. C., Jäggi, L., Hinckley, K., Castellanos, A., **Aguilar, L.** et al. *Digital support systems to improve child development in Peru: A cluster-randomized controlled open-label trial*. Science Advances 12(10), eaeb9403.

2024 **Aguilar, L.**, Gath-Morad, M., Grübel, J., Ermatinger, J., Zhao, H., Wehrli, S. et al. *Experiments as Code and its application to VR studies in human-building interaction*. Scientific Reports 14(1), 9883.

2021 **Aguilar, L.**, Dao, D., Gan, S., Gürel, N. M., Hollenstein, N., Jiang, J. et al. *Ease.ML: A Lifecycle Management System for MLDev and MLOps*. 11th Annual Conference on Innovative Data Systems Research (CIDR).

2019 **Aguilar, L.**, Bennati, S., Helbing, D. *How learning can change the course of evolution*. PLoS ONE 14(9), e0219502.

2018 Wijerathne, L., Petprakob, W., **Aguilar, L.**, Hori, M., Ichimura, T. *Scalable HPC enhanced agent based system for simulating mixed-mode evacuation of large urban areas*. Transportation Research Procedia 34, 275–282.

2017 **Aguilar, L.**, Lalith, M., Ichimura, T., Hori, M. *On the performance and scalability of an HPC-enhanced Multi-Agent System based evacuation simulator*. Procedia Computer Science 108, 937–947.

Full list: [Google Scholar](#).

Grants & Funding

2012–2015 **Ph.D. Scholarship**, MEXT (Japan), The University of Tokyo. *Main applicant* (competitive: 2 national scholarships). Subject: scalable multi-agent large-urban-area evacuation simulator.

- 2012–2015 **HPC enhancement of an emergency evacuation simulator**, JSPS, The University of Tokyo. *Ph.D. researcher* (competitive).
- 2013–2017 **Assessment of Earthquake Disasters for Next-Generation Urban Area Models**, JSPS, The University of Tokyo. *Postdoctoral researcher* (competitive).
- 2013–2018 **MOMENTUM**, ERC Advanced Grant (no. 324247), ETH Zurich. *Postdoctoral researcher*.
- 2016–2020 **The Anatomy of Systemic Financial Risk**, SNSF Interdisciplinary Projects (no. 162776), ETH Zurich. *Postdoctoral researcher*.
- 2018 **Predicting Biases** (Planning Grant), Volkswagen Foundation (**AI and the Society of the Future**), ETH Zurich. *Co-awardee*: coordinated the consortium and wrote the proposal.
- 2020–2025 **CoCi: Co-Evolving City Life**, ERC (no. 833168), ETH Zurich: contributed to the proposal writing.
- 2025–2029 **EVIDENT: Evidence-Based Architectural Design of Emergency Departments**, SNSF International Co-Investigator Scheme, ETH Zurich. *Project partner* and contributed to the proposal writing.

Awards & Scholarships

- 2024 **Guatemalteco Ilustre — Orator**, Seguros Universales, Guatemala. [Invited talk at the award event](#).
- 2019 **MIT Technology Review Innovator Under 35, Latin America** for large-scale human-mobility simulation (10M simulated evacuees across Tokyo).
- 2018 Planning grant, **VW Foundation**, AI and its impacts on tomorrow's world.
- 2017 National Center for Research on Earthquake Engineering, **IDEER's Award**.
- 2012 **Japanese Government MEXT Scholarship** (full Ph.D. scholarship).
- 2011–2012 UMALCA scholarships for EMALCA (Latin-American & Caribbean Mathematics Schools): El Salvador & Mexico (2011), Venezuela (2011), and Costa Rica (*Differential Geometry* diploma, 2012).
- Honours *Magna Cum Laude* (M.Sc., USAC, 2011); *Cum Laude* (B.Sc., UVG, 2009); multiple Outstanding Student Awards, UVG (2004, 2005, 2007, 2008).

Participation in Research Projects

- BRCCH **Digital Support Systems to Improve Child Health & Development in Low-Income Settings**, AI virtual assistant deployed in Peru (cluster-randomized trial, *Science Advances* 2026).
- ERC **MOMENTUM**, ERC Advanced Investigator Grant (EU).
- SNSF **The Anatomy of Systemic Financial Risk**, Swiss National Science Foundation.
- Post-K Disaster Prevention & Environmental Problems, Priority Issue #3 (Japan, RIKEN/U-Tokyo).
- Industry MeteoSwiss (terabyte-scale AutoML for weather forecasting); Swiss Data Science Center (AutoML on the Renku platform).

Invited Lectures, Service & Outreach

- 2026 Contributor, **Behavioral Clones Workshop 2026**, Max Planck Institute for Human Development, Berlin.
- 2020 **Invited lecture**: “An Interactive Introduction to Data Science for Engineers”, INSPIRE Training School, ETH Zurich.
- 2018 **Chair**: 4th International Workshop on Data-driven Self-regulating Systems (DSS 2018).
- 2017–2024 Multiple invited lectures, **CONVERCIENCIA**, Guatemala; talks at major Guatemalan universities and the engineering association.
- 2019–2024 **Chief Academic Advisor**, GuateML (machine-learning education initiative, Guatemala).
- 2015 **Organizer**: Japan–Malaysia Student Exchange Workshop.
- 2012 **Organizer**: 3rd Japan–Sino Workshop on Massive Simulation Systems for Earthquake Disaster.
- Reviewing *Simulation Modelling Practice and Theory*; *International Journal of Disaster Risk Reduction*; ACM Collective Intelligence; EUSPN (2018–2019); IEEE/ACM UCC Doctoral Research Forum (2018).

Media Coverage & Invited Talks

- 2024 *Guatemalteco Ilustre* orator talk series; TV / radio / podcast interviews, Guatemala.
- 2020 **Forbes Latin America**, **Fortune en Español**, **MIT Technology Review** (Spanish), **Innovators Under 35** coverage. **MIT TR IU35 LATAM pitch** on large-scale human-behaviour simulation.
- 2017–2024 Invited talks at Guatemala's scientific meetings and major universities; multiple TV and radio interviews.
- 2012–2017 Newspaper features (*Prensa Libre*, *Nuestro Diario*).